

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Criteria	Understanding of Scenario (1)	Identification of Risks / Issues (1)	Application of Concepts (1.25)	Decision Justification (1)	Clarity (0.75)	TOTAL (5)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Scenario-Based Questions

1. A financial institution plans to deploy a private cloud with strict compliance requirements. Recommend an appropriate cloud datacenter architecture and explain how virtualization and isolation mechanisms can ensure security and regulatory compliance.
2. A SaaS provider experiences inconsistent application performance due to noisy neighbor effects in a virtualized environment. Analyze the root cause and propose resource allocation and hypervisor-level solutions.
3. An organization wants to migrate legacy monolithic applications into a Service-Oriented Architecture (SOA) in the cloud. Explain the migration strategy and how SOA improves scalability and maintainability.
4. A government agency is implementing identity federation across multiple departments using cloud services. Explain how identity management, authentication, and privacy policies should be designed to ensure secure access.
5. A cloud provider wants to implement real-time user presence detection for collaborative applications. Discuss how presence services can be integrated into cloud architecture while maintaining user privacy and system scalability.

1. A financial institution plans to deploy a private cloud with strict compliance requirements. Recommend an appropriate cloud datacenter architecture and explain how virtualization and isolation mechanisms can ensure security and regulatory compliance.

Solution

A **Private Cloud Datacenter** is the most suitable architecture for a financial institution because it provides complete control over infrastructure, security, and regulatory compliance. The datacenter should include physical servers, enterprise storage (SAN/NAS), high-speed networking, a virtualization layer using hypervisors (VMware ESXi, Hyper-V, or KVM), a cloud management platform (OpenStack or VMware vCloud), security services, and banking applications.

Virtualization enables multiple Virtual Machines (VMs) to run independently on the same physical server while sharing hardware resources efficiently. Each VM has its own operating system, memory, storage, and applications, preventing interference between workloads. Isolation mechanisms such as VM isolation, network isolation (VLAN/VXLAN), storage isolation, and resource isolation ensure that applications remain secure and independent. Hypervisors allocate dedicated CPU, memory, storage, and network resources to each VM, preventing unauthorized access and resource contention.

To ensure regulatory compliance, the cloud environment should implement Identity and Access Management (IAM), Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA), AES-256 encryption for stored data, TLS 1.3 encryption for data transmission, audit logging, continuous monitoring through SIEM tools, regular backups, disaster recovery, and compliance with standards such as PCI DSS, ISO 27001, and GDPR. These mechanisms provide confidentiality, integrity, availability, and accountability while meeting financial industry regulations.

2. A SaaS provider experiences inconsistent application performance due to noisy neighbor effects in a virtualized environment. Analyze the root cause and propose resource allocation and hypervisor-level solutions.

Solution

The inconsistent performance experienced by the SaaS provider is caused by the **Noisy Neighbor Effect**, where multiple virtual machines share the same physical resources. If one VM consumes excessive CPU,

memory, disk I/O, or network bandwidth, neighboring VMs experience slower performance, increased response times, and higher latency.

The root causes include CPU contention, memory contention, storage I/O bottlenecks, and network congestion. Since all virtual machines share the underlying hardware, excessive resource usage by one application negatively affects others running on the same host.

The problem can be solved through effective resource allocation. CPU and memory reservations guarantee minimum resources for critical applications, while storage Quality of Service (QoS) limits disk input/output operations per VM. Network QoS reserves bandwidth for important applications, preventing congestion. Resource pools can also be created to separate production, testing, and development environments.

At the hypervisor level, fair CPU scheduling, NUMA-aware memory allocation, live migration (such as VMware vMotion), storage migration, and dynamic load balancing help distribute workloads evenly across physical hosts. Continuous monitoring using tools like VMware vCenter, Prometheus, or Grafana enables administrators to detect resource bottlenecks early and optimize resource allocation. These measures ensure consistent application performance, improved scalability, and better user experience.

3. An organization wants to migrate legacy monolithic applications into a Service-Oriented Architecture (SOA) in the cloud. Explain the migration strategy and how SOA improves scalability and maintainability.

Solution

The migration of a legacy monolithic application to a **Service-Oriented Architecture (SOA)** should begin by analyzing the existing application to identify business functions, dependencies, databases, and communication patterns. The monolithic application should then be divided into independent services such as authentication, customer management, payment processing, inventory, and notification services.

Each service should expose standardized APIs using REST or SOAP protocols for communication. Depending on the application requirements, services may use separate databases or a shared database, although separate databases are generally preferred to reduce dependencies. The services can then be deployed independently in cloud environments using Virtual Machines or containers managed by Kubernetes.

Comprehensive unit testing, integration testing, performance testing, and security testing should be performed before production deployment. Monitoring tools such as Prometheus, Grafana, and the ELK Stack can be used to monitor the health and performance of each service.

SOA improves scalability because only heavily utilized services need to be scaled instead of the entire application. It also enhances maintainability by allowing developers to update, test, and deploy individual services independently without affecting the entire system. The architecture provides better fault isolation, improved code reusability, easier integration with third-party applications, and greater flexibility for future enhancements.

4. A government agency is implementing identity federation across multiple departments using cloud services. Explain how identity management, authentication, and privacy policies should be designed to ensure secure access.

Solution

A government agency implementing identity federation across multiple departments should establish a centralized **Identity Provider (IdP)** that manages user identities and authentication for all participating departments. Identity federation allows users to authenticate once and securely access multiple cloud applications using **Single Sign-On (SSO)**.

Identity management should use **Role-Based Access Control (RBAC)** to assign permissions according to user roles such as administrator, officer, employee, or citizen. Authentication should be strengthened using **Multi-Factor Authentication (MFA)**, combining passwords with one-time passwords (OTP) or biometric verification.

Industry-standard federation protocols such as **SAML 2.0**, **OAuth 2.0**, and **OpenID Connect (OIDC)** should be implemented to enable secure authentication and authorization across multiple departments.

Privacy policies should focus on collecting only necessary user information, encrypting data both at rest and in transit, obtaining user consent where applicable, defining data retention policies, and maintaining detailed audit logs. Security measures such as encryption, least privilege access, continuous monitoring, regular security audits, and incident response planning help protect sensitive government information while ensuring compliance with privacy regulations. This architecture provides secure, centralized, and seamless access to cloud services across different government departments.

5. A cloud provider wants to implement real-time user presence detection for collaborative applications. Discuss how presence services can be integrated into cloud architecture while maintaining user privacy and system scalability.

Solution

A cloud provider can implement **real-time user presence detection** by integrating a dedicated presence service into the cloud architecture. The architecture should include client applications, an API gateway, authentication services, a presence server, distributed cache (such as Redis), message brokers (Kafka or RabbitMQ), databases, and notification services using WebSockets or MQTT.

When users log into the application, they are authenticated before periodic heartbeat messages update their online status. The presence server stores this information in a distributed cache and instantly broadcasts status changes to authorized users through the notification service. This enables real-time collaboration in messaging, video conferencing, and document editing applications.

To protect user privacy, authentication and authorization mechanisms should ensure that only authorized users can view presence information. Users should be able to control their visibility status, including options such as **Online, Busy, Away, Invisible, or Offline**. All communication should be encrypted using TLS, while stored presence information should be protected using AES-256 encryption. Only essential presence data should be stored, and comprehensive audit logs should be maintained.

Scalability can be achieved through horizontal scaling of presence servers, load balancing across multiple instances, distributed caching with Redis clusters, asynchronous message processing using Kafka or RabbitMQ, and independent deployment of presence services as microservices. Continuous monitoring ensures efficient resource utilization and rapid detection of failures. This architecture delivers low-latency presence updates while maintaining high availability, strong privacy protection, and support for millions of concurrent users.